Rutherford's Atomic Model

Physics • Grade 11

Student Name: _____ **Date:** _____ **Score:** ___ / 5

Read each item carefully and answer.

Part A: Choose the right answer

Tick one box.

1. Which scientist proposed the Rutherford's atomic model?

- ☐ **A.** Niels Bohr
- ☐ **B.** Ernest Rutherford
- ☐ **C.** James Chadwick
- ☐ **D.** J.J. Thomson

2. What did Rutherford's model predict about the path of most alpha particles when they hit the gold foil?

- ☐ **A.** They would be absorbed by the foil.
- ☐ **B.** They would pass straight through without deflection.
- ☐ **C.** They would be deflected at large angles.
- ☐ **D.** They would be deflected at small angles.

Part B: True or false

Tick True or False.

3. True or False: The Rutherford's model suggests that the atom is mostly empty space with a dense nucleus at the center.

☐ **True** ☐ **False**

4. True or False: Rutherford's model explains the stability of the atom by the presence of equal numbers of protons and electrons.

☐ **True** ☐ **False**

Part C: Fill in the blank

Choose a word from the word bank.

5. In Rutherford's experiment, a beam of _____ was directed at a thin gold foil.

Word bank: **protons alpha particles neutrons electrons**

Answer Key

Teacher copy • Rutherford's Atomic Model

1. A. Niels Bohr — Ernest Rutherford was the scientist who conducted the gold foil experiment and proposed the nuclear model of the atom.

2. C. They would be deflected at large angles. — Most alpha particles passed straight through the foil, indicating that the atom is mostly empty space.

3. True — Rutherford's model showed that the atom has a small, dense nucleus surrounded by electrons, making the majority of the atom's volume empty space.

4. False — Rutherford's model did not explain the stability of the atom in terms of the number of protons and electrons, but rather the structure of the atom with a dense nucleus and orbiting electrons.

5. alpha particles — Rutherford used a beam of positively charged alpha particles in his experiment to test the structure of the atom.